Technical Report: Resistive Bridge Measurements

Wiring, Signal Sensing, and Lead-Resistance Compensation

June 10, 2026

1 Instrumentation Core Concepts

A resistive sensor reports a measurand (temperature, strain, force, or a similar quantity) as a change in electrical resistance. To be recorded, this resistance must first be converted into a voltage. The conversion is performed by an instrument, such as a datalogger, a dedicated resistance-thermometer meter, or a multimeter, which supplies the electrical excitation and contains the voltmeter that constitutes its analog input. The sensor, of resistance R_s, is generally located at the end of a cable whose conductors each possess a finite resistance, denoted $R_{L1}, R_{L2}, \ldots$. Because this lead resistance is added to the quantity of interest, its treatment governs the accuracy of the measurement and motivates the different wiring configurations examined in this report.

2 Measurement principle

Two excitation schemes are in common use.

- **Voltage Source with Completion Resistor (Ratiometric Method):** The instrument applies a stable voltage (V_X) across a circuit containing an internal, fixed completion resistor (R_f). Together, R_f and the sensor form a voltage divider. The measurement is recorded as a ratio:

  $$X = \frac{V_{\text{out}}}{V_X}$$

  Any baseline drift in V_X cancels out because it affects both the sensed voltage and the reference voltage simultaneously.

- **Constant-Current Source:** The instrument drives a known, constant current (I) directly through the sensor. The resulting voltage drop across the sensor is measured to calculate resistance ($R_s = V/I$). This topology requires no completion resistor; the current remains stable regardless of changes in R_s.

The term *bridge* carries two distinct meanings, which are kept separate throughout. As a circuit topology, a half bridge is a two-element voltage divider, whereas a full bridge is a four-arm Wheatstone network. In the terminology of strain measurement, the descriptors quarter, half, and full denote the number of active gauges among the four arms of a Wheatstone bridge; all three are measured as full bridges.

3 Half-bridge configuration (Voltage Divider)

In the ratiometric half bridge, the completion resistor R_f and the sensor R_s are connected in series across the excitation, and the voltmeter measures the potential at their junction (Figure 1). The transfer relation is

$$X = \frac{V_{\text{out}}}{V_X} = \frac{R_s}{R_f + R_s}, \qquad R_s = R_f\,\frac{X}{1 - X}. \tag{1}$$

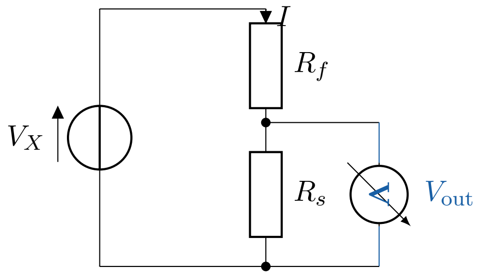

Figure 1: Ratiometric half bridge. The completion resistor R_f and the excitation V_X reside in the instrument; the sensor resistance is obtained from the ratio of V_{out} to V_X through Equation (1).

This configuration is appropriate for a single resistive sensor, such as a thermistor, a single resistance thermometer, or a potentiometer. It provides no compensation for the resistance of the connecting cable, which is the subject of Section 4.

4 Lead-resistance compensation: two-, three-, and four-wire connections

Each conductor of the cable contributes a lead resistance. The accuracy of the measurement depends on whether the voltmeter senses a voltage that includes this lead resistance. In the figures that follow, each connection is shown for both excitation schemes of Section 2: case (a), a voltage source with completion resistor R_f; and case (b), a constant-current source. The dashed boundary encloses the instrument, which contains the source and the voltmeter; the cable and sensor lie outside it. The excitation current flows only through the current-carrying leads, shown in red. The voltmeter is treated as ideal, drawing no current, so a lead that carries only the voltmeter connection develops no voltage drop.

4.1 Two-wire connection

With two conductors, the voltmeter shares them with the current path (Figure 2). The sensed voltage therefore spans $R_{L1} + R_s + R_{L2}$, and the lead resistances cannot be distinguished from the sensor. This connection is acceptable when the sensor resistance is large and the cable is short, and is unsuitable for low-resistance sensors or long cables.

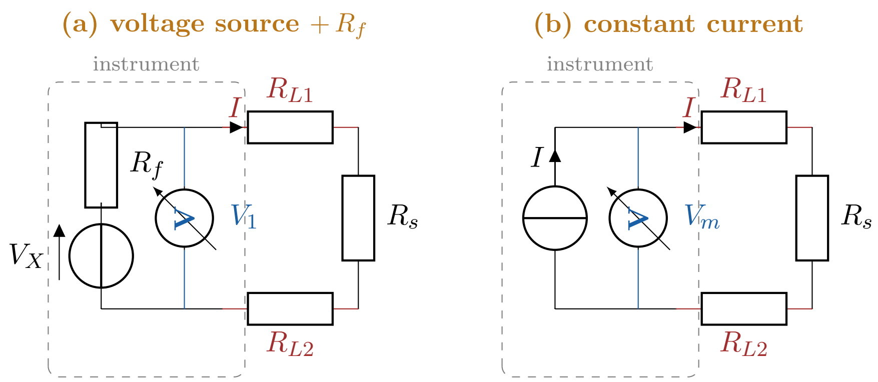

Figure 2: Two-wire connection. In both schemes the voltmeter shares the current-carrying leads, so the sensed voltage corresponds to $R_{L1} + R_s + R_{L2}$. Scheme (a) reports V_1/V_X; scheme (b) reports V_m/I.

4.2 Three-wire connection

A third conductor, R_{L2}, is connected to the upper terminal of the sensor but carries no current, as it serves only the voltmeter; it therefore senses that terminal directly (Figure 3). Voltage V_1 spans the complete loop, $R_{L1} + R_s + R_{L3}$, whereas voltage V_2 excludes the supply lead and spans $R_s + R_{L3}$. The combination $2V_2 - V_1$ removes the lead contribution when the leads are matched in gauge, length, and temperature. This connection renders low-resistance sensors, such as a $100\,\Omega$ platinum element, usable over practical cable lengths.

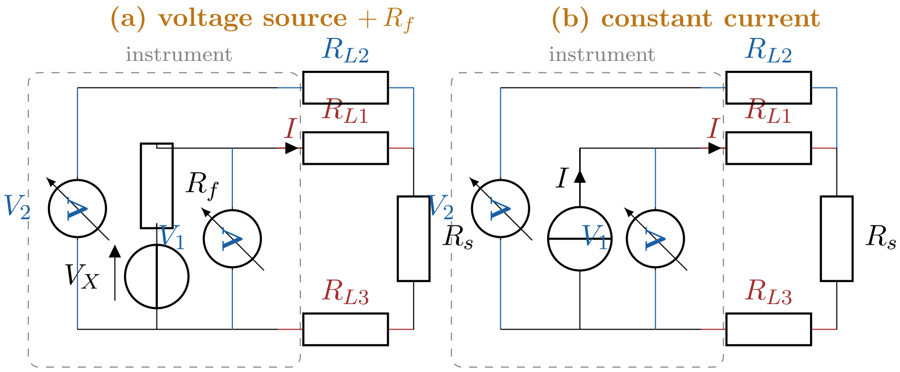

Figure 3: Three-wire connection. The two schemes differ only in the source block. The sense lead R_{L2} (upper, blue) carries no current, and the correction $2V_2 - V_1$ is identical in both cases.

4.3 Four-wire connection

In the four-wire (Kelvin) connection, the current leads, R_{L1} and R_{L2}, are separate from the sense leads, R_{L3} and R_{L4} (Figure 4). The excitation current flows only in the force pair, while the sense pair is connected to the terminals of the sensor and carries no current; its lead resistance therefore develops no voltage drop, and the voltmeter records the true voltage across the sensor irrespective of cable length. In scheme (a) the sensor resistance is obtained as the ratio $R_s = R_f\,(V_s/V_f)$, where V_f is measured directly across the local completion resistor; in scheme (b) it is $R_s = V_m/I$.

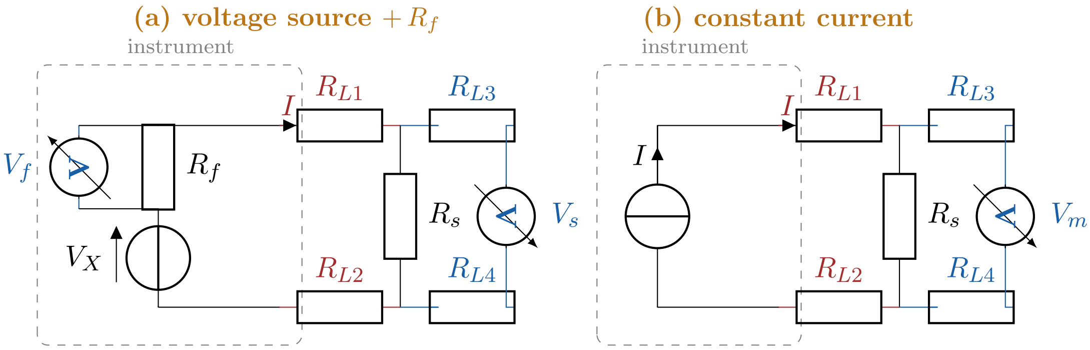

Figure 4: Four-wire (Kelvin) connection. The sense pair carries no current, so V_s (scheme a) or V_m (scheme b) equals the true voltage across R_s for any cable length. Scheme (a) additionally measures V_f across the local R_f to form the ratio.

5 Determination of the sensor resistance and the role of the excitation

In the ratiometric scheme, the excitation current is not imposed but is determined by the loop, $I = V_X/(R_f + R_{L1} + R_s + R_{L3})$. The denominator contains the unknown sensor and lead resistances, so the current is not obtained from this expression. It is instead obtained from the voltage developed across the completion resistor R_f, whose value is known. Since the upper terminal of R_f is at the excitation potential V_X and its lower terminal is the node measured as V_1, the drop across R_f is $V_X - V_1$, and

$$I = \frac{V_X - V_1}{R_f}. \tag{2}$$

The quantity $V_X - (2V_2 - V_1)$ does not give this drop: it equals $I\,(R_f + R_{L1} + R_{L3})$ and therefore includes the lead resistances, so its use in place of $V_X - V_1$ overstates the drop across R_f.

Combining the loop voltages with Equation (2) yields the sensor resistance for each connection of the ratiometric scheme:

$$\text{two-wire:}\quad R_s + R_{L1} + R_{L2} = R_f\,\frac{V_1}{V_X - V_1}, \tag{3}$$

$$\text{three-wire:}\quad R_s = R_f\,\frac{2V_2 - V_1}{V_X - V_1}, \tag{4}$$

$$\text{four-wire:}\quad R_s = R_f\,\frac{V_s}{V_f}. \tag{5}$$

The two- and three-wire connections sense voltages only with respect to the instrument ground; the drop across R_f is consequently inferred as $V_X - V_1$, and the value of V_X is required, as shown by Equations (3) and (4). The four-wire connection measures V_f directly across R_f, so V_X does not appear in Equation (5).

Ratiometric cancellation. The requirement to know V_X does not imply that V_X must be an accurately calibrated voltage. The same excitation is used both to drive the divider and as the reference of the analog-to-digital converter, so the measured quantity is the ratio V_1/V_X. Any drift or absolute error in V_X appears identically in the sensed voltage and in the reference, and therefore cancels; only the ratio is required. The four-wire connection does not form a ratio against V_X at all.

As a numerical verification of Equations (2) and (4), consider the three-wire connection with $V_X = 2.5\,\mathrm{V}$, $R_f = 1000\,\Omega$, $R_s = 100\,\Omega$, and $R_{L1} = R_{L2} = R_{L3} = 10\,\Omega$. The loop current is $I = 2.232\,\mathrm{mA}$, giving $V_1 = 0.2679\,\mathrm{V}$ and $V_2 = 0.2455\,\mathrm{V}$. Then $2V_2 - V_1 = 0.2232\,\mathrm{V} = IR_s$ and $V_X - V_1 = 2.232\,\mathrm{V} = IR_f$, so $R_s = 1000 \times 0.2232/2.232 = 100\,\Omega$, in agreement with the assumed value.

For the constant-current scheme there is no excitation voltage and no completion resistor. The sensor resistance follows from the known current: $R_s + R_{L1} + R_{L2} = V_m/I$ for the two-wire connection, $R_s = (2V_2 - V_1)/I$ for the three-wire connection, and $R_s = V_s/I$ for the four-wire connection. In this scheme the accuracy depends on the calibration of the current source rather than on V_X.

6 Error analysis

For the two-wire connection the measured resistance is $R_{\text{meas}} = R_s + R_{L1} + R_{L2}$, equal to $R_s + 2R_L$ for matched leads, so the fractional error is $2R_L/R_s$. The significance of this error depends on the nominal sensor resistance, as summarised in Table 1.

A $100\,\Omega$ platinum element has a temperature coefficient of approximately $0.385\,\Omega\,{}^\circ\mathrm{C}^{-1}$, so a lead resistance of $10\,\Omega$ corresponds to a temperature error of about $26\,{}^\circ\mathrm{C}$. The same lead resistance applied to a $10\,\mathrm{k}\Omega$ thermistor produces a fractional error of $0.1\,\%$. Accordingly, a high-resistance sensor on a short cable may be measured with two wires; a low-resistance sensor, or any sensor on a long cable, requires the three-wire connection; and the four-wire connection is used when the lead error must be eliminated entirely.

Table 1: Two-wire lead-resistance error for a representative $2R_L = 10\,\Omega$.

Sensor	nominal R_s	fractional error	approximate temperature error
Thermistor	$10\,\mathrm{k\Omega}$	0.1%	negligible
$1000\,\Omega$ RTD	$1000\,\Omega$	1%	$\sim 2.5\,°\mathrm{C}$
$100\,\Omega$ RTD	$100\,\Omega$	10%	$\sim 26\,°\mathrm{C}$

7 Full-bridge configuration

A full bridge consists of two voltage dividers connected in parallel (Figure 5). The excitation is applied across one diagonal, and the voltmeter measures the difference of potential across the other diagonal:

$$X = \frac{V_{\text{out}}}{V_X} = \frac{R_1}{R_1 + R_3} - \frac{R_2}{R_2 + R_4}. \tag{6}$$

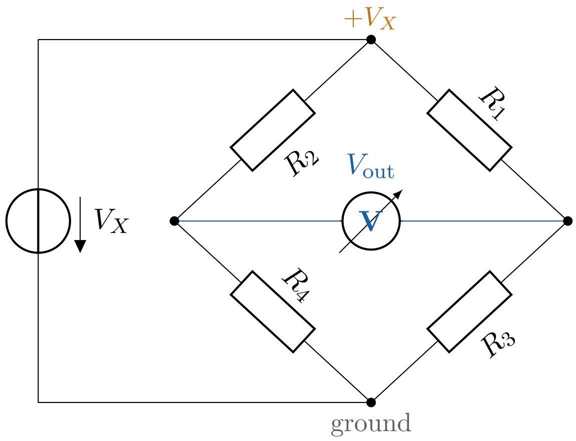

Figure 5: Four-arm Wheatstone (full) bridge. The excitation is applied across one diagonal and the voltmeter across the other; a balanced bridge produces zero output.

The full bridge is employed where the fractional change in resistance is small and both sensitivity and the rejection of common effects are required, as in strain gauges, load cells, and pressure and force transducers. Because the output is a difference, a balanced bridge produces a small signal that may be measured on a high-resolution range, and because the four arms experience a common temperature and excitation, effects common to all arms cancel in the difference. A six-wire variant adds two conductors that sense the excitation at the bridge terminals, thereby removing the voltage drop along the excitation cable on long connections.

8 Strain-gauge configurations

In strain measurement, the descriptors quarter, half, and full denote the number of active gauges among the four arms of a Wheatstone bridge; the remaining arms are fixed resistors (Figure 6). All three are measured as full bridges.

The quarter bridge, comprising one active gauge and three fixed completion resistors, is the simplest and most widely used arrangement. The half bridge may be configured to reject bending or temperature, according to the orientation of the second gauge. The full bridge produces the largest output and the best inherent compensation, which is the reason that commercial load cells are constructed internally as full bridges.

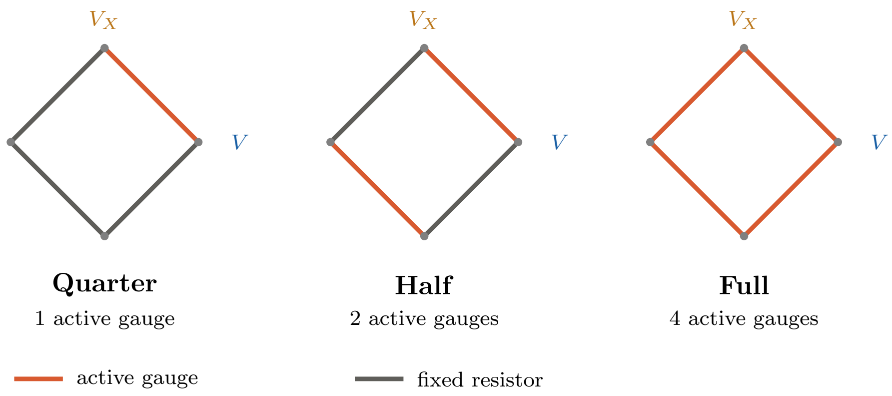

Figure 6: Quarter-, half-, and full-bridge strain-gauge arrangements. A larger number of active arms yields a larger output for a given strain and improves the cancellation of temperature and bending effects.

9 Configuration selection

Table 2 summarises the configurations and their typical applications.

Table 2: Selection of measurement configuration.

Configuration	Topology	Typical application
Two-wire	divider	thermistor, potentiometer, short cable
Three-wire	divider	low-resistance sensor (e.g. 100 Ω RTD), moderate cable
Four-wire	divider / direct	resistance thermometer, long cable, maximum lead rejection
Full bridge	Wheatstone	strain gauge, load cell, pressure transducer
Six-wire full bridge	Wheatstone	as above, long cable or highest accuracy

10 Practical considerations

Sensors immersed in or coupled to conductive media, including electrolytic tilt sensors, soil-moisture blocks, water-conductivity cells, and wetness grids, as well as inductive sensors such as linear variable differential transformers, require an alternating (AC) excitation in which the polarity is reversed on successive readings. Continuous single-polarity excitation of such sensors causes polarisation, calibration drift, and accelerated degradation. Most measurement instruments provide an excitation-reversal option for this purpose.

11 Conclusion

The configuration of a resistive-sensor measurement determines whether the resistance of the connecting cable corrupts the result. The two-wire connection includes the lead resistance in a single reading; the three-wire connection cancels matched lead resistance by combining two readings as $2V_2 - V_1$; and the four-wire connection eliminates lead resistance by sensing the voltage across the sensor with conductors that carry no current. In the ratiometric scheme the two- and three-wire connections require the excitation voltage in order to determine the loop current from the completion resistor, whereas the four-wire connection does not; in all cases the use of a common excitation reference causes errors in V_X to cancel. The full bridge extends these principles to four active or partially active arms for the measurement of small resistance changes with inherent common-mode rejection.